Introduction to Database Systems (CSD202)

Lab Assignment#9

Date: 05/04/19

You have to upload the screen shot (black screen) of output with the queries one by one, in a word file & file name must be-> (Roll No) _Assignment#8.

******Work under DBMS2019 database*****

```
mysql> use dbms2019;
Database changed
```

Create Two Tables Employee & Log_Details as per given below data & declaration.

Table 1:Employee

Emp_ID	Int(5), Primary Key
Name	Varchar(30)
Age	Int(3)
Adult	int

Table 2: Log_Details

Emp_ID	Int(5)
Name	Varchar(30)
Edit_DateTime	DateTime, not null

```
mysql> create table Employee
    -> (
    -> Emp_ID int(5) PRIMARY KEY,
    -> Name varchar(30),
    -> Age int (3),
    -> Adult int);
Query OK, 0 rows affected (0.07 sec)

mysql> create table Log_Details
    -> (
    -> Emp_ID int(5),
    -> Name varchar(30),
    -> Edit_DateTime DateTime NOT NULL);
Query OK, 0 rows affected (0.05 sec)

mysql> describe employee;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| Emp_ID | int(5)      | NO   | PRI | NULL    |       |
| Name   | varchar(30) | YES  |     | NULL    |       |
| Age    | int(3)      | YES  |     | NULL    |       |
| Adult  | int(11)     | YES  |     | NULL    |       |
+--------+-------------+------+-----+---------+-------+
4 rows in set (0.03 sec)

mysql> describe log_details;
+---------------+-------------+------+-----+---------+-------+
| Field         | Type        | Null | Key | Default | Extra |
+---------------+-------------+------+-----+---------+-------+
| Emp_ID        | int(5)      | YES  |     | NULL    |       |
| Name          | varchar(30) | YES  |     | NULL    |       |
| Edit_DateTime | datetime    | NO   |     | NULL    |       |
+---------------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

<u>Problem 1</u>: Create a Before insert trigger **Adult_check** for table employee table. The logic of this trigger is, when you insert a record to the table employee and if employee age is <18 years then adult column will be auto filled by 0 else by 1. Means you will not insert the data in adult column it will be auto filed from age column.

Table 1:Employee

Emp_ID	Emp_Name	Age	Adult
10001	ABC	22	
10002	DEF	15	
10003	GHI	12	
10004	JKL	28	

Table 2: Log_Details

Emp_ID	Name	Edit_Timde
10001	ABC	Current date time
10002	DEF	Current date time
10003	GHI	Current date time
10004	JKL	Current date time

```
mysql> delimiter //
mysql> create trigger Adult_Check
    -> before insert on employee
    -> for each row
    -> begin
    -> if new.age<18 then set new.adult = 0;
    -> else set new.adult = 1;
    -> end if;
    -> end;
    -> //
Query OK, 0 rows affected (0.03 sec)

mysql> insert into employee (Emp_ID, Name, Age) values (10001,'ABC',22)//
Query OK, 1 row affected (0.02 sec)

mysql> insert into employee (Emp_ID, Name, Age) values (10002,'DEF',15)//
Query OK, 1 row affected (0.01 sec)

mysql> insert into employee (Emp_ID, Name, Age) values (10003,'GHI',12)//
Query OK, 1 row affected (0.01 sec)

mysql> insert into employee (Emp_ID, Name, Age) values (10004,'JKL',28)//
Query OK, 1 row affected (0.01 sec)

mysql> select * from employee//
+--------+------+------+-------+
| Emp_ID | Name | Age  | Adult |
+--------+------+------+-------+
|  10001 | ABC  |   22 |     1 |
|  10002 | DEF  |   15 |     0 |
|  10003 | GHI  |   12 |     0 |
|  10004 | JKL  |   28 |     1 |
+--------+------+------+-------+
4 rows in set (0.00 sec)
```

<u>Problem 2:</u> Create & execute after insert trigger **New_record** on table employee, when you insert the data to employee table, then same data (Emp_ID & Name) will be auto inserted to table Log_Details with edit time. Means you will insert the data in employee table and log_details will be automatically updated.

```
mysql> create trigger New_Record
    -> after insert on employee
    -> for each row
    -> begin
    -> insert into log_details values (New.emp_ID, new.name, NOW());
    -> end;
    -> //
Query OK, 0 rows affected (0.02 sec)

mysql> insert into employee (Emp_ID, Name, Age) values (10001,'ABC',22)//
ERROR 1062 (23000): Duplicate entry '10001' for key 'PRIMARY'
mysql> insert into employee (Emp_ID, Name, Age) values (10002,'DEF',15)//
ERROR 1062 (23000): Duplicate entry '10002' for key 'PRIMARY'
mysql> delete from employee where 1=1;
    -> //
Query OK, 4 rows affected (0.02 sec)

mysql> insert into employee (Emp_ID, Name, Age) values (10001,'ABC',22)//
Query OK, 1 row affected (0.01 sec)

mysql> insert into employee (Emp_ID, Name, Age) values (10002,'DEF',15)//
Query OK, 1 row affected (0.01 sec)

mysql> insert into employee (Emp_ID, Name, Age) values (10003,'GHI',12)//
Query OK, 1 row affected (0.01 sec)

mysql> insert into employee (Emp_ID, Name, Age) values (10004,'JKL',28)//
Query OK, 1 row affected (0.01 sec)

mysql> select * from log_details//
+--------+------+---------------------+
| Emp_ID | Name | Edit_DateTime       |
+--------+------+---------------------+
|  10001 | ABC  | 2019-04-06 18:33:25 |
|  10002 | DEF  | 2019-04-06 18:33:25 |
|  10003 | GHI  | 2019-04-06 18:33:26 |
|  10004 | JKL  | 2019-04-06 18:33:26 |
+--------+------+---------------------+
4 rows in set (0.00 sec)
```

<u>Problem 3:</u> Create & execute after update trigger **Update_record** on table Employee, if you want to update any record in employee table, then changes will be reflecting in Log_Details table. Means you will update only employee table & Log_Details will be updated automatically.

```
mysql> create trigger update_record
    -> after update on employee
    -> for each row
    -> begin
    -> update Log_details
    -> set name = new.name where emp_ID=old.emp_id;
    -> end;
    -> //
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> update employee
    -> set name = 'Kabir' where name='ABC';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from employee;
+--------+-------+------+-------+
| Emp_ID | Name  | Age  | Adult |
+--------+-------+------+-------+
|  10001 | Kabir |   22 |     1 |
|  10002 | DEF   |   15 |     0 |
|  10003 | GHI   |   12 |     0 |
|  10004 | JKL   |   28 |     1 |
+--------+-------+------+-------+
4 rows in set (0.00 sec)

mysql> select * from log_details;
+--------+-------+---------------------+
| Emp_ID | Name  | Edit_DateTime       |
+--------+-------+---------------------+
|  10001 | Kabir | 2019-04-06 18:33:25 |
|  10002 | DEF   | 2019-04-06 18:33:25 |
|  10003 | GHI   | 2019-04-06 18:33:26 |
|  10004 | JKL   | 2019-04-06 18:33:26 |
+--------+-------+---------------------+
4 rows in set (0.00 sec)
```

<u>Problem 4:</u> Create & execute after delete trigger **Delete_record** on table employee. If you delete any record from employee, then this record will be deleted from log_details. Means you will delete the record from the table employee & same record will be deleted from Log_Details automatically.

```
mysql> delimiter //
mysql> create trigger delete_record
    -> after delete on employee
    -> for each row
    -> begin
    -> delete from log_details where emp_id = old.emp_id;
    -> end;
    -> //
Query OK, 0 rows affected (0.01 sec)

mysql> delete from employee where name='Kabir'//
Query OK, 1 row affected (0.01 sec)

mysql> select * from employee;
    -> //
+--------+------+------+-------+
| Emp_ID | Name | Age  | Adult |
+--------+------+------+-------+
|  10002 | DEF  |   15 |     0 |
|  10003 | GHI  |   12 |     0 |
|  10004 | JKL  |   28 |     1 |
+--------+------+------+-------+
3 rows in set (0.00 sec)

mysql> select * from log_details//
+--------+------+---------------------+
| Emp_ID | Name | Edit_DateTime       |
+--------+------+---------------------+
|  10002 | DEF  | 2019-04-06 18:33:25 |
|  10003 | GHI  | 2019-04-06 18:33:26 |
|  10004 | JKL  | 2019-04-06 18:33:26 |
+--------+------+---------------------+
3 rows in set (0.00 sec)
```

***implement before update & before delete triggers in free time**

What free time?

<u>Problem 5:</u> Display all the triggers in DBMS2019 & Drop Them.

```
mysql> show triggers;
    -> //
+---------------+--------+----------+-----------------------------------------------------------
-------------------+--------+-----------------------+-----------------------------------------
-+----------------+----------------------+----------------------+--------------------+
| Trigger       | Event  | Table    | Statement
                    | Timing | Created                 | sql_mode
 | Definer        | character_set_client | collation_connection | Database Collation |
+---------------+--------+----------+-----------------------------------------------------------
-------------------+--------+-----------------------+-----------------------------------------
-+----------------+----------------------+----------------------+--------------------+
| Adult_Check   | INSERT | employee | begin
if new.age<18 then set new.adult = 0;
else set new.adult = 1;
end if;
end | BEFORE | 2019-04-06 18:28:47.14 | STRICT_TRANS_TABLES,NO_ENGINE_SUBSTITUTION | root@localhos
t | cp850                | cp850_general_ci     | utf8mb4_0900_ai_ci |
| New_Record    | INSERT | employee | begin
insert into log_details values (New.emp_ID, new.name, NOW());
end         | AFTER  | 2019-04-06 18:32:56.94 | STRICT_TRANS_TABLES,NO_ENGINE_SUBSTITUTION | root@
localhost | cp850                | cp850_general_ci     | utf8mb4_0900_ai_ci |
| update_record | UPDATE | employee | begin
update Log_details
set name = new.name where emp_ID=old.emp_id;
end       | AFTER  | 2019-04-06 18:36:53.43 | STRICT_TRANS_TABLES,NO_ENGINE_SUBSTITUTION | root@lo
calhost | cp850                | cp850_general_ci     | utf8mb4_0900_ai_ci |
| delete_record | DELETE | employee | begin
delete from log_details where emp_id = old.emp_id;
end                   | AFTER  | 2019-04-06 18:42:02.04 | STRICT_TRANS_TABLES,NO_ENGINE_SUBSTITUT
ION | root@localhost | cp850                | cp850_general_ci     | utf8mb4_0900_ai_ci |
+---------------+--------+----------+-----------------------------------------------------------
-------------------+--------+-----------------------+-----------------------------------------
-+----------------+----------------------+----------------------+--------------------+
4 rows in set (0.01 sec)
```

```
mysql> drop trigger adult_check//
Query OK, 0 rows affected (0.01 sec)

mysql> drop trigger new_record//
Query OK, 0 rows affected (0.02 sec)

mysql> drop trigger update_record//
Query OK, 0 rows affected (0.02 sec)

mysql> drop trigger delete_record//
Query OK, 0 rows affected (0.01 sec)
```

“Drop dbms2019”